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Draw

Formulas

Data

Review

View



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Enter text or formula here



	A	B	C	D	E	F	G	H	I	J
1	v1	v2								
2	ham	Hello how are you								
3	spam	Congratulations you won a free iPhone								
4	ham	Let's meet tomorrow								
5	spam	Claim your prize now								
6	ham	See you at college								
7	spam	You have won 10000 rupees								
8	ham	Call me when you arrive								
9	spam	Click here to get free gifts								
10	ham	Good morning								
11	spam	Limited time offer buy now								
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[15]

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```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
data = pd.read_csv("/spam.csv")
data = data[['v1', 'v2']]
data.columns = ['label', 'message']
data['label'] = data['label'].map({'ham': 0, 'spam': 1 })
X_train, X_test, y_train, y_test = train_test_split(data[
vectorizer = CountVectorizer()
X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_features, y_train)
predictions = model.predict(X_test_features)
accuracy = accuracy_score(y_test, predictions)
print("Model Accuracy:", accuracy)
sample_email = ["this is to inform you to join meeting imr
sample_features = vectorizer.transform(sample_email)
result = model.predict(sample_features)
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```

... Model Accuracy: 1.0
Spam Email





Run all cells in notebook



[12]

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```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
ata = pd.read_csv("/spam.csv")
ata = ata[['v1', 'v2']]
ata.columns = ['label', 'message']
ata['label'] = ata['label'].map({'ham': 0, 'spam': 1})
X_train, X_test, y_train, y_test = train_test_split(ata['message'],
                                                    ata['label'])
vectorizer = CountVectorizer()
X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_features, y_train)
predictions = model.predict(X_test_features)
accuracy = accuracy_score(y_test, predictions)
print("Model Accuracy:", accuracy)
sample_email = ["how are you"]
sample_features = vectorizer.transform(sample_email)
result = model.predict(sample_features)
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```



... Model Accuracy: 1.0
Not Spam





Run all cells in notebook



[10]

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```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
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X_train, X_test, y_train, y_test = train_test_split(data['message'],
                                                    data['label'])
vectorizer = CountVectorizer()
X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_features, y_train)
predictions = model.predict(X_test_features)
accuracy = accuracy_score(y_test, predictions)
print("Model Accuracy:", accuracy)
sample_email = ["good morning"]
sample_features = vectorizer.transform(sample_email)
result = model.predict(sample_features)
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```

... Model Accuracy: 1.0
Not Spam





[7]

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
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X_train, X_test, y_train, y_test = train_test_split(data['message'], data['label'])
vectorizer = CountVectorizer()
X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_features, y_train)
predictions = model.predict(X_test_features)
accuracy = accuracy_score(y_test, predictions)
print("Model Accuracy:", accuracy)
sample_email = ["Congratulations! You have won a free iPhone"]
sample_features = vectorizer.transform(sample_email)
result = model.predict(sample_features)
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```

... Model Accuracy: 1.0
Spam Email





Run all cells in notebook



[9]

0s



```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
data = pd.read_csv("/spam.csv")
data = data[['v1', 'v2']]
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data['label'] = data['label'].map({'ham': 0, 'spam': 1})
X_train, X_test, y_train, y_test = train_test_split(data['message'], data['label'])
vectorizer = CountVectorizer()
X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_features, y_train)
predictions = model.predict(X_test_features)
accuracy = accuracy_score(y_test, predictions)
print("Model Accuracy:", accuracy)
sample_email = ["Congratulations! You got a chance to win"]
sample_features = vectorizer.transform(sample_email)
result = model.predict(sample_features)
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```

... Model Accuracy: 1.0
Not Spam

